

**CSE490 (D)**

**Introduction to  
Football Data & Analytics**

**Lecture-11**

**Pass Probability Modeling**

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# Context

- Last lecture: pitch control; who controls each region of the field.
- This lecture: pass probability=> given a player kicks the ball, what is the chance a teammate receives it?
- Paper: Spearman et al. (2017), MIT Sloan Sports Analytics Conference.
- Self read: the paper itself (uploaded with the syllabus).

# The Problem

- A pass involves: 1 passer, 1 ball, 21 other players — all moving.
- Outcome depends on ball trajectory, defenders, receivers, skill, decision-making.
- We want a single number:  $P(\text{pass succeeds})$  — before the ball is kicked.
- This number unlocks: scouting, tactical analysis, pass valuation, decision evaluation.

# Two Existing Approaches

## Time-to-intercept methods

- Compute how long each player needs to reach the ball.
- Strength: physically grounded.
- Weakness: rigid, no probabilities.

## Machine learning methods

- Train a classifier on hand-engineered features.
- Strength: flexible, fits anything.
- Weakness: black box, hard to interpret.

*This paper: a hybrid — physics-derived inputs into a small statistical model with interpretable parameters.*

# Four Design Principles

- Probabilistic — output a real number in  $[0, 1]$ .
- Data-driven — fit to real games, not assumed.
- Predictive — uses only information available at the moment of the pass.
- Smooth — similar situations get similar probabilities.

*Smoothness is load-bearing — it's what makes the pitch control function continuous, not Voronoi-like.*

# Two Conditions for a Successful Reception

A player receives a pass if and only if both of these happen:

## 1. Time to Intercept

Reach the ball's path before it moves past you.

*Modeled with the Logistic CDF.*

## 2. Time to Control

Stay near the ball long enough to bring it under control.

*Modeled with the Exponential CDF.*

# Step 1: The Ball's Trajectory

The ball obeys an equation of motion:

$$\ddot{\mathbf{r}} = -g \hat{\mathbf{z}} - (1 / 2m) \cdot \rho \cdot C_D \cdot A \cdot \dot{\mathbf{r}} \cdot |\dot{\mathbf{r}}|$$

Three forces in real life: gravity, drag, Magnus (spin). Paper keeps gravity + drag, drops Magnus.

| Symbol            | Value                  |
|-------------------|------------------------|
| g (gravity)       | 9.8 m/s <sup>2</sup>   |
| m (ball mass)     | 0.42 kg                |
| ρ (air density)   | 1.22 kg/m <sup>3</sup> |
| C_D (drag)        | 0.25                   |
| A (cross-section) | 0.038 m <sup>2</sup>   |

## Two simplifications:

- Ball travels in a straight line (no curve)
- Drag still applies on the ground (no rolling friction)

## Step 2: Time to Intercept (Setup)

For each point along the ball's trajectory, ask: how long would this player take to reach that point?

Player's equation of motion (constant acceleration):

$$r = r_i + \dot{r} \cdot t + (1/2) \cdot \ddot{r} \cdot t^2$$

*Constraints:  $|\text{velocity}| \leq v_{\text{max}}$ ,  $|\text{acceleration}| \leq a_{\text{max}}$*

Solve for the smallest  $t$  such that the player reaches the ball at flight time  $T$ .

**Call this  $t_{\text{int}}(T)$  — the player's time to intercept.**



## Step 2: Time to Intercept (Probability)

**Define spare time:  $\Delta t = T - t_{\text{int}}$**

If  $\Delta t \geq 0$ , the player can reach. But there's noise: reaction time, fatigue, footing, awareness.

Model the uncertainty with the Logistic CDF:

$$P_{\text{int}}(T) = 1 / [ 1 + \exp( -(T - t_{\text{int}}) / s ) ] \quad \text{where } s = \text{sqrt}(3) \cdot \sigma / \pi$$

- $\sigma$  — standard deviation of arrival-time uncertainty (free parameter, fit from data)
- Big positive  $\Delta t \rightarrow P \approx 1$  (player makes it)
- Big negative  $\Delta t \rightarrow P \approx 0$  (can't reach)
- $\Delta t \approx 0 \rightarrow P = 0.5$  (toss-up)

## Step 3: Time to Control

- Reaching the ball  $\neq$  controlling it.
- A defender might brush the ball with a foot but fail to win possession.
- A clean trap takes time.

**Introduce the control rate  $\lambda$ :**

$$\textit{Probability of gaining control during a tiny slice } \Delta t = \lambda \cdot \Delta t$$

This is a constant-rate assumption, same idea as radioactive decay or Poisson processes. Each instant near the ball is an independent chance to gain control.

# Step 3: Time to Control (Probability)

Integrating the constant rate gives the Exponential CDF:

$$P(t) = 1 - e^{(-\lambda t)}$$

Plugging in  $\lambda = 4.30 \text{ s}^{-1}$  (the fitted value):

| Time near ball (s) | P(controlled) |
|--------------------|---------------|
| 0.1                | 35%           |
| 0.25               | 66%           |
| 0.5                | 88%           |
| 1.0                | 99%           |
| 2.0                | 99.98%        |

*Mean time to control:  $1/\lambda \approx 0.23 \text{ s}$ . Control happens fast or not at all.*

# Combining the Two (Equation 5)

Rate at which player  $j$  wins the ball at flight time  $T$ :

$$\frac{dP_j}{dT}(T) = \left(1 - \sum_k P_k(T)\right) P_{\text{int},j}(T) \lambda$$

Three factors multiplied:

- $(1 - \sum P_k)$  — probability the ball is still in play (no other player has won it yet)
- $P_{\text{int},j}$  — probability  $j$  is in proximity right now (interception piece)
- $\lambda$  — rate at which proximity converts to control (exponential rate)

*The recursion is what makes the model realistic — players further along the path lose probability to closer ones.*

# Total Probability and Pass Success

Integrate over all flight times:

$$P_j(t) = \int_0^t \left( 1 - \sum_k P_k(T) \right) P_{\text{int}}(T) \lambda \, dT$$

For a successful pass, sum over teammates of the passer (set A):

$$P_A = \sum_{j \in A} P_j$$

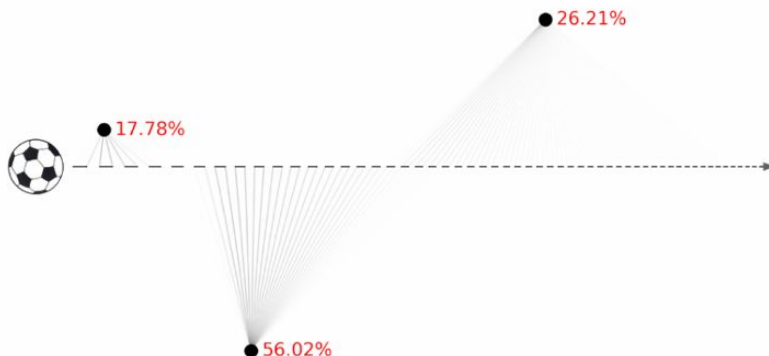
Computed numerically — discrete time-stepping until probabilities saturate ( $t \rightarrow \infty$  in practice).

*This is a horse race in time: at every instant, redistribute remaining probability among players who can intercept.*

# Worked Example (Paper Figure 3)

Three players along a ball's trajectory:

| Player | Position          | P(receive) | Why?                                     |
|--------|-------------------|------------|------------------------------------------|
| 1      | Close to passer   | 17.8%      | Ball still moves too fast here           |
| 2      | Middle            | 56.0%      | Best position , multiple intercept lines |
| 3      | Far down the path | 26.2%      | Earlier players already won the ball     |



# The Dataset

- 38 games of Crystal Palace, 2015-16 Premier League.
- Tracking data at 25 fps (every player + ball, every frame).
- Event data (passes, tackles, goals, etc.).
- Filtering: passes 0.5–10 s long, low curvature, in-bounds → 10,875 passes.
  - Training: 5,404 passes
  - Test: 5,471 passes
  - All 38 games used for downstream applications

## Receiver definition:

the next player after the passer to do one of {pass, tackle, claim, ball recovery, keeper pick-up}.

# Bernoulli Framing

Each pass is a coin flip with bias  $p$ :

$$\begin{aligned} P(k \mid \sigma, \lambda, x) &= 1 - p && \text{for } k = 0 \quad (\text{failure}) \\ P(k \mid \sigma, \lambda, x) &= p && \text{for } k = 1 \quad (\text{success}) \end{aligned}$$

For a single pass with known outcome and game state, the likelihood of  $(\sigma, \lambda)$  is:

$$L(\sigma, \lambda \mid k, x) = P(k \mid \sigma, \lambda, x)$$

*Same formula, flipped perspective , fix outcome, ask which parameters are most plausible.*



# Maximum Likelihood Estimation

Find  $(\sigma, \lambda)$  that maximize total likelihood over all 5,404 training passes.

Equivalently: minimize negative log-likelihood.

$$\min_{\sigma, \lambda \in \{\mathbb{R}, \mathbb{R}\}} \left\{ - \sum_{i \in P} \log[\mathcal{L}(\sigma, \lambda | k_i, x_i)] \right\}$$

- Why log? Multiplying 5,000 probabilities  $\rightarrow$  numerical underflow. Log turns products into sums.
- Why negative? Most optimization tools minimize. Same answer either way.

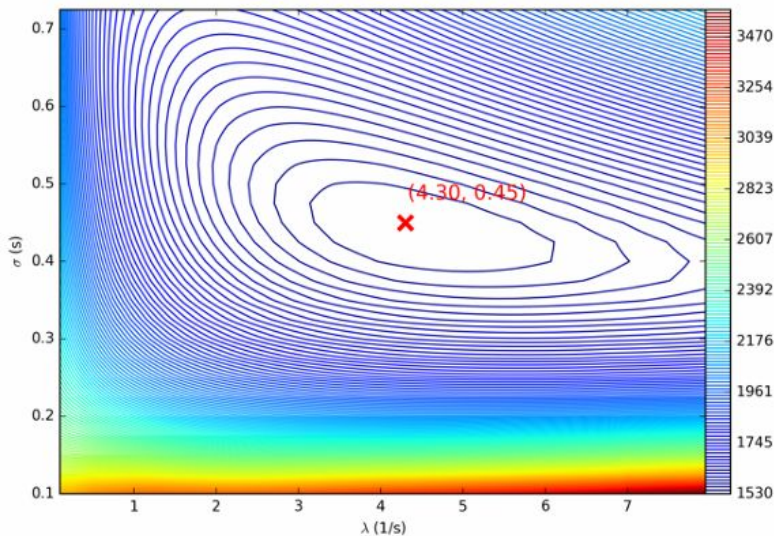
*This is the same recipe that trains logistic regression and neural networks. The model is unusual; the fitting is textbook.*

# Grid Search Results

Only two parameters  $\rightarrow$  no need for fancy optimization. Just try every  $(\sigma, \lambda)$  pair on a grid.

$$\sigma = 0.45 \pm 0.04 \text{ s}$$

$$\lambda = 4.30 \pm 1.1 \text{ s}^{-1}$$



- Likelihood surface has a single sharp minimum  $\rightarrow$  fit is statistically stable.
- $\pm$  values are systematic uncertainty (re-fit per game, std deviation across games).

# Do These Numbers Make Sense?

$$\sigma = 0.45 \text{ s}$$

*Interception time uncertainty*

- 95% of arrivals within  $\pm 0.9$  s of computed  $t_{\text{int}}$
- Plausible given human variability in reaction, effort, footing

$$\lambda = 4.30 \text{ s}^{-1}$$

*Control rate*

- Mean time to control:  $1/\lambda \approx 0.23$  s
- 99% control within 1 second of proximity
- Matches intuition: trapping is fast for elite players

# Headline Accuracy

Evaluated on 12 held-out games (5,471 test passes):

| Metric                                   | Value |
|------------------------------------------|-------|
| Predicting receiving team                | 80.5% |
| Predicting specific player               | 67.9% |
| With cutoff tuned to 0.27 instead of 0.5 | 81.9% |
| ROC AUC                                  | 0.85  |

**With only two parameters.**

*Most ML pass-prediction models use dozens of features. This one uses physics + 2 numbers.*

# The Calibration Gap

**78.9%**

Actual completion rate

**67.9%**

Average predicted completion

**Model is systematically pessimistic by ~11 points. Why?**

| Missing factor                | Direction of bias                    |
|-------------------------------|--------------------------------------|
| Magnus force (curving passes) | Real passes safer than model thinks  |
| Player skill differences      | Real players better than average     |
| Tracking noise                | Adds spurious uncertainty            |
| Strategy / pass selection     | Real passes pre-filtered for success |

# Reception & Interception Efficiency

For each player: expected receptions vs. actual receptions = efficiency ratio.

| Position   | Total | Receiving | Interception |
|------------|-------|-----------|--------------|
| Defender   | 1.06  | 1.04      | 1.13         |
| Midfielder | 1.03  | 1.23      | 0.57         |
| Forward    | 0.91  | 1.36      | 0.23         |
| Goalkeeper | 0.65  | 0.80      | 0.48         |

*Forwards are best at receiving (1.36) but worst at intercepting (0.23). This is role-driven, not skill-driven.*

# Pass Value

Combine probability with positional gain:

$$V_j = p_j f(x_{\text{suc}}) - (1 - p_j) f(x_{\text{fail}})$$

$f(x)$  = value of being at position  $x$ . Paper uses a simple exponential of distance to goal.

## Logic:

- High-probability pass to a low-value position has small value.
- Low-probability pass to a high-value position can still be worth attempting.

*Same idea as expected goals (xG) but for passes. Separates decision quality from execution quality.*

# Correlations with League Standing

For each of 20 EPL teams, compute mean reception efficiency and pass value across Crystal Palace games.

| Metric                       | Corr. with shots | Corr. with attacking-third passes |
|------------------------------|------------------|-----------------------------------|
| Reception efficiency (total) | 0.64             | 0.70                              |
| Pass value (mean)            | 0.63             | 0.83                              |

**0.83 correlation is striking.**

A metric derived from physics + 2 parameters predicts attacking-third activity better than most hand-engineered features.



# Extension 1: Pitch Control Function

What if we computed pass probability for an imaginary ball at every point on the pitch?

→ **The Pitch Control Function (PCF): a heatmap of who controls what.**

**Use case: defending a corner kick.**

| Within 5m of goal | Defending team's PCF |
|-------------------|----------------------|
| Shot is saved     | 0.82                 |
| Shot scores       | 0.79                 |
| Difference        | 0.04                 |

*Defending the central area really does matter. Quantitative confirmation of conventional wisdom.*

# Extension 2: Hypothetical Passing

Replace the actual ball velocity with an imagined one. Ask: what's the best pass from here?

1. Use simulated annealing to find optimal velocity  $v_{\text{ideal}}$  that maximizes  $P(\text{pass to target receiver})$ .
2. Perturb  $v_{\text{ideal}}$  slightly to get a distribution of probabilities  $P$ .
3. Report two numbers:  $\mu_P + \sigma_P$  (well-kicked) and  $\mu_P - \sigma_P$  (poorly-kicked).

**Reading the two numbers:**

- Both high  $\rightarrow$  easy pass
- Both low  $\rightarrow$  impossible pass
- High top / low bottom  $\rightarrow$  feasible but requires skill

# What's Elegant About This Paper

- Two parameters do the work of fifty — physics structures the function space.
- Interpretable —  $\sigma$  and  $\lambda$  have units and meaning, not just weights in a matrix.
- Smooth by construction — adjacent situations get adjacent probabilities, no cliff edges.
- Composable — same model gives pass probability, pitch control, hypothetical passing.
- Honest about limits — names every assumption it breaks.

*Compare to a 200-feature gradient-boosted model with similar accuracy. Numbers are similar; understanding is not.*

# What the Paper Gets Wrong

| Limitation                                   | Fix today                                 |
|----------------------------------------------|-------------------------------------------|
| No Magnus force                              | Add ball spin from camera tracking        |
| Uniform $\sigma$ , $\lambda$ for all players | Per-player parameters; hierarchical model |
| Straight-line ball trajectory                | 3D ball physics with spin                 |
| No team strategy prior                       | Bayesian prior conditioned on game state  |
| Slow simulated annealing                     | Neural surrogate for ideal-pass search    |

*Every weakness here is a research project. That's the mark of a foundational paper — it doesn't close the field, it opens it.*

# How the Field Has Built on This

- Spearman 2018: extended to off-ball scoring opportunities (xT-style metrics).
- Fernández & Bornn 2018: convex pitch control with explicit defensive pressure.
- Laurie Shaw's "Friends of Tracking": open-source reproductions on Metrica data.
- Liu et al. 2020+: neural network pitch control replacing the analytical model.
- Modern pro clubs: pitch control is now a standard tactical analysis layer.

*The 2017 paper is the seed. Almost every modern football analytics paper using tracking data cites it.*

# Insights for Your Own Modeling Work

- Use physics where you can; fit parameters for what you can't. Hybrids beat extremes.
- Few interpretable parameters > many opaque ones, when accuracy is comparable.
- Honest error analysis matters more than headline numbers.
- A good model spawns extensions for free. Same physics → PCF, pass value, hypotheticals.

*Lessons that transfer beyond football.*